Rebound and static settling

PFC3D 7.0 official example | One Md=0, beta_n=0.6 subcase | 29 October 2026

PASS	All four runs compared with the published absolute-coefficient target; this is not a full 32-ball reproduction.

First-rebound height comparison

Run	Apex-derived e	Absolute error	Rel. error (%)	Apex z (m)	Target
CPU dt	0.0881547443	0.0066255	6.99036	0.032576977	PASS
GPU dt	0.0881547443	0.0066255	6.99036	0.032576977	PASS
CPU dt/2	0.0888625165	0.0059177	6.24361	0.032699133	PASS
GPU dt/2	0.0888625165	0.0059177	6.24361	0.032699133	PASS

e0 = exp[-beta_n pi/sqrt(1-beta_n^2)] = 0.094780224842. e_height = sqrt[(z_apex-R)/(z0-R)], first rebound only. Published target: abs(e_height-e0) < 0.01. FISH calculates absolute coefficient difference; relative error is diagnostic only. This is not a relative 1% tolerance or automated assertion.

Second contact and static equilibrium

Run	Final center z (m)	Final Fn (N)	Late max z error	Late max speed
CPU dt	0.024965311581	1.734420944	6.6e-17	9.1e-15
GPU dt	0.024965311581	1.734420944	6.6e-17	9.1e-15
CPU dt/2	0.024965311581	1.734420944	2.2e-16	8.8e-14
GPU dt/2	0.024965311581	1.734420944	2.2e-16	8.8e-14

Targets: delta_eq=mg/kn=3.468841888e-05 m; z_eq=0.024965311581 m; Fn_eq=mg=1.7344209442 N. Declared supplemental checks over 0.8-1.0 s: height error <1e-9 m, relative force error <1e-6, speed <1e-8 m/s. These are not official PFC limits. The second contact persists to 1 s.

Official inputs and native mapping

Quantity	Retained official value / exact active mapping
Ball 12 / gravity	R=0.025 m; rho=2650 kg/m^3; center=(1.15,0.05,1) m; initial velocity/spin zero. Fixed floor z=0; g=-10 m/s^2 throughout; free-fall clearance=0.975 m.
Normal law / damping	kn=50000 N/m; beta_n=0.6; Md=0. e_input=0.094780224842; automatic cn=111.7489633 N s/m; full effective mass=0.1734420944 kg. Manual dashpots off/zero; other contact channels and local/global damping zero.
Time / outputs	dt=10/5 us; 100000/200000 numerical steps to 1 s. Every-step CSV; native frames every 1 ms, 1001/run, 4004 total. Four loadable projects and complete results retained.

One original subcase; full signed dashpot

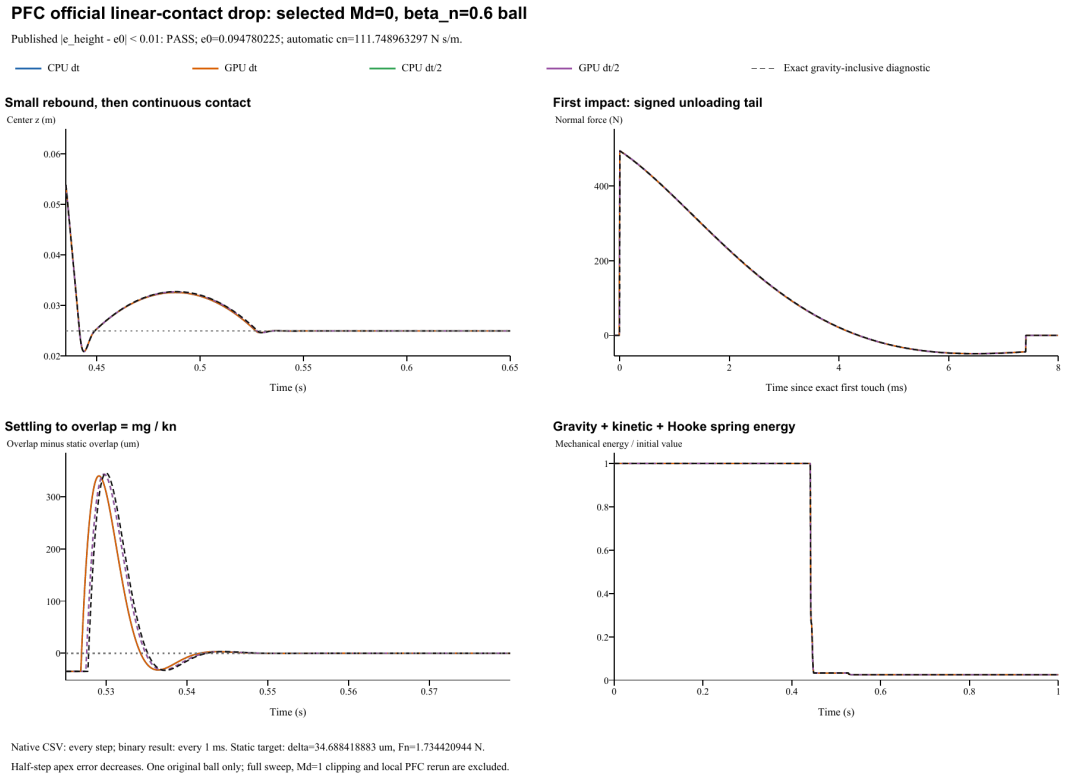
Original ball 12 only; no full 32-ball sweep or Md=1 clipping. Md=0 retains signed viscous unloading, not adhesion. No coefficient was fitted or solver changed. The previous beta_n=0.2 case remains unchanged.

Gravity and reference provenance

The analytic e0 relation neglects gravity during contact. FunDEM keeps gravity active; an independent gravity-inclusive solution gives e_height=0.0891422438. A finite contact-duration offset can survive refinement. Only the first rebound is measured, using every-step CSV rather than 1 ms viewing frames.

Official PFC7 example and embedded input | Source/contact audit.
Versioned documentation web-connector text snapshot SHA-256: 3dc3744ccb9cc04fb3ec1db4a2d44f27f83b50b8e187020b03c54ab80f00e7e5. PFC was not rerun locally; no Git commit/blob is invented.

Native histories and retained evidence



Automatic normal damping and wall impulse

Run	Inferred cn	Min. Fn (N)	Force residual	Impulse residual
CPU dt	111.748963297	-48.90583	1.4e-14	4.2e-13
GPU dt	111.748963297	-48.90583	5.7e-14	4.2e-13
CPU dt/2	111.748963297	-49.01194	2.8e-14	2.1e-12
GPU dt/2	111.748963297	-49.01194	5.7e-14	2.1e-12

Force timing includes gravity: vn_assembly = vz_saved - (dt/2)(Fn/m-10); cn_inferred=(kn delta-Fn)/vn_assembly. The audit excludes near-zero denominators and checks direct force reconstruction. Coefficients are inferred from native state, not a direct output channel.

CPU/GPU whole-history maximum position / velocity / signed-force differences are 1.11e-16 m / 2.71e-14 m/s / 1.83e-11 N.

Halving dt changes apex-derived restitution by 0.0007077722. The absolute differences from the published analytic coefficient are 0.006625481 / 0.005917708. The half-step error decreases for both the published and gravity-inclusive references.

The fixed wall supplies external impulse, while gravity acts throughout. The maximum residual in m(v-v0) = integral(F_contact + mg)dt is 2.07e-12 kg m/s. Sphere momentum alone is not conserved.

Mechanical energy includes gravity and spring energy once; the dashpot dissipates energy. Energy conservation and particle momentum conservation are not acceptance conditions here. Tangential force, torque and spin remain inactive.

Complete results and scope

All 4004 configured native frames, run/project manifests, four input projects, every-step particle/contact CSV and CPU/CUDA logs are retained. Real SimulationSession executes linked FunDEM 0.4.3 native libraries. Installed GUI executable byte identity is not established.

Run manifest | Analysis JSON | Independent audit | Complete native ZIP | Frame/hash audit

Verified scope: one official Md=0 stronger-damping subcase, first rebound, persistent second contact, static settling and gravity-aware force/impulse integration. The full 32-ball sweep, Md=1 clipping and active friction/rotation remain outside scope. Prior beta_n=0.2 results remain unchanged; PFC follows numbered LAMMPS demos in the overview.